2.3 #s: 3, 5, 8, 9

2.4 #s: 1, 3, 4, 9

Homework 4, Math 3210. By: Tyler Trotter

2.3 3: Use the Main Limit Theorem to find $\lim \left(\frac{2^n}{2^n+1}\right)$.

Lemma: $\lim \frac{1}{2^n} = 0$ whenever $n > N = \log_2\left(\frac{1}{\varepsilon}\right)$ for any ε.

$$\lim\left(\frac{2^n}{2^n+1} \cdot \frac{1/2^n}{1/2^n}\right) = \lim\left(\frac{1}{1+\frac{1}{2^n}}\right) \overset{d}{=} \frac{\lim(1)}{\lim\left(1+\frac{1}{2^n}\right)} \overset{b}{=} \frac{\lim(1)}{\lim(1)+\lim\left(\frac{1}{2^n}\right)} \overset{a}{=} \frac{1}{1+\lim\left(\frac{1}{2^n}\right)}$$

$$\overset{\text{Lemma}}{=} \frac{1}{1+0} = \frac{1}{1} = \boxed{1}.$$

5: Prove Theorem 2.3.2. Let $\{a_n\}$ be a sequence of real numbers such that $\lim a_n = 0$, and let $\{b_n\}$ be a bounded sequence. Then $\lim a_n b_n = 0$.

Since b_n is bounded, $\forall n$, $M < b_n < k$, then $|b_n| < \max\{M, k\}$ $\neq 0$

Claim $\lim a_n b_n = 0$.

Proof: Let $\varepsilon > 0$. Since $\lim a_n = 0$, $\exists N \in \mathbb{R}$ s.t. $\forall n > N$, $|a_n - 0| = |a_n| < \frac{\varepsilon}{\max\{M, k\}}$. It follows that for the same N, if $n > N$, then $|a_n b_n - 0| = |a_n b_n| = |a_n||b_n| < \frac{\varepsilon}{\max\{M,k\}} \cdot \max\{M, k\} = \varepsilon$.

By def 2.1.4 $\lim a_n b_n = 0$. $\blacksquare$

2.3 #s: 8, 9
2.4 #s: 1, 3, 4, 9

Homework 4, cont.

2.3 #8: Prove that if $\{b_n\}$ is a sequence of positive terms and $b_n \to b > 0$, then there is a number $m > 0$ such that $b_n \geq m$ for all n.

By Corallary 2.2.4, all convergent sequences are bounded (and definition of bounded meaning there exists an $m \leq b_n$ for all b_n).

Proof:
Since, by corallary b_n must be bounded, there is an m less than or equal to all b_n; $m \leq b_n \ \forall n$.
Since b_n is a sequence of positive values that converges to some value $b > 0$, the sequence has only finitely many elements for which $|b_n - b| \geq \varepsilon$.
What this means is that there are not infinitely many elements of b_n more than ε distance from b.
Thus, there are only a finite number of elements between (0) and $(\varepsilon$ from $b)$. Thus there must exist a number m that is not 0 but is less than or equal to any b_n. $\blacksquare$

9: Claim $\lim \frac{a_n}{b_n} = \frac{a}{b}$.
Proof: Since $\lim b_n = b \neq 0$, $\lim |b_n| = |b| > 0$. By Exercise 8, $\exists m > 0 : |b_n| > m \ \forall n \in \mathbb{N}$. This means $\frac{1}{|b_n|} < \frac{1}{m} \ \forall n$.
Now let $\varepsilon > 0$. Since $\lim a_n = a$, $\exists N_1 \in \mathbb{R}$ s.t. $\forall n > N_1$, $|a_n - a| < \frac{m\varepsilon}{2}$.
Since $\lim b_n = b$, $\exists N_2 \in \mathbb{R}$ s.t. $\forall n > N_2$: $|b_n - b| < \frac{|b|}{|a|} \cdot \frac{m\varepsilon}{2}$.
Let $N = \max\{N_1, N_2\}$. If $n > N$ then:

$$\left|\frac{a_n}{b_n} - \frac{a}{b}\right| = \left|\frac{ba_n - ab_n}{bb_n}\right| = \left|\frac{ba_n - ab + ab - ab_n}{bb_n}\right| = \left|\frac{b(a_n - a) - a(b_n - b)}{bb_n}\right|$$

$$\underset{\Delta \text{ ineq}}{\leq} \left|\frac{b(a_n - a)}{bb_n}\right| + \left|\frac{-a(b_n - b)}{bb_n}\right| = \frac{|a_n - a|}{|b_n|} + \frac{|a|}{|b|}\frac{|b_n - b|}{|b_n|} < \frac{1}{m} \cdot \frac{m\varepsilon}{2} + \frac{|a|}{|b|}\left(\frac{1}{m}\frac{|b|}{|a|} \cdot \frac{m\varepsilon}{2}\right) = \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Thus, $\lim \frac{a_n}{b_n} = \frac{a}{b}$. $\blacksquare$

2.4 #s: 1, 3, 4, 9

Homework 4, cont.

2.4 #1: a) $\{n^2\}$ is non-decreasing, thus $a_n \leq a_{n+1}$.

Claim: $n^2 \leq (n+1)^2$ for all $n \in \mathbb{N}$.

Base Case: $(1)^2 \leq (1+1)^2$; $1 \leq 4$ is true.

Inductive Step: Assume $n^2 \leq (n+1)^2$ for all n. Proceed by induction to show $(n+1)^2 \leq (n+2)^2$.

$$(n+1)^2 = n^2 + 2n + 1 < n^2 + 4n + 4 = (n+2)^2$$

Thus, $(n+1)^2 \leq (n+2)^2$ for all n and we have to prove $n^2 \leq (n+1)^2$ for all n. ■

b) $\left\{\frac{1}{\sqrt{n}}\right\}$ is non-increasing, thus $a_n \geq a_{n+1}$.

Claim: $\frac{1}{\sqrt{n}} \geq \frac{1}{\sqrt{n+1}}$ for all $n \in \mathbb{N}$.

Base Case: $\frac{1}{\sqrt{1}} \geq \frac{1}{\sqrt{2}}$; $1 < \sqrt{2}$, so $1 \geq \frac{1}{\sqrt{2}}$ is true.

Inductive Step: Assume $\frac{1}{\sqrt{n}} \geq \frac{1}{\sqrt{n+1}}$ is true. Then proceed by induction to show $\frac{1}{\sqrt{n+1}} \geq \frac{1}{\sqrt{n+2}}$ is true.

$\frac{1}{\sqrt{n+1}} \geq \frac{1}{\sqrt{n+2}}$ is trivial to show: $\sqrt{n+1} < \sqrt{n+2}$ for any $n \in \mathbb{N}$. ■

$\left\{\frac{1}{\sqrt{n}}\right\}$ is also bounded because 1 is its largest element because it is non-increasing. It is also bounded below because there is no $n \in \mathbb{N}$ s.t. $\frac{1}{\sqrt{n}}$ is negative or even 0.

c) $\left\{\frac{(-1)^n}{n}\right\}$ is bounded because there is no element $n \in \mathbb{N}$ s.t.

a) $\frac{(-1)^n}{n} < -1$ and b) $\frac{1}{2} < \frac{(-1)^n}{n}$.

Proof. a) $\frac{(-1)^n}{n} \geq -1$ $\forall n \in \mathbb{N}$.

Base Case $\frac{(-1)^1}{1} \geq -1$; $-1 = -1$ is true.

Inductive Step: Assume $\frac{(-1)^n}{n} \geq -1$ is true for all n; $(-1)\frac{(-1)^n}{n} < (-1)(-1)$

$$\frac{(-1)^{n+1}}{n+1} = \frac{(-1)(-1)^n}{n+1} < (-1)\frac{(-1)^n}{n} < (-1)(-1) = 1.$$

True.

b) $\frac{1}{2} \geq \frac{(-1)^n}{n}$ $\forall n \in \mathbb{N}$. Base Case: $\frac{(-1)^1}{1} \leq \frac{1}{2}$ is true

Inductive Step: Assume $\frac{(-1)^n}{n} \leq \frac{1}{2}$ $\forall n \in \mathbb{N}$. need to show $\frac{(-1)^{n+1}}{n+1} < \frac{1}{2}$.

$$\frac{(-1)(-1)^n}{n+1} < (-1)\frac{(-1)^n}{n} \leq (-1)\frac{1}{2} \leq \frac{1}{2}.$$

Homework 4, cont

2.4 #1 continued

d) $\left\{\frac{n}{2^n}\right\} = \left(\frac{1}{2}, \frac{1}{2}, \frac{3}{8}, \frac{1}{4}, \ldots\right)$ is bounded & non-increasing.

No value for n will yield a negative or even 0.
Likewise $1/2 \geq \frac{n}{2^n}$ $\forall n \in \mathbb{N}$.
Need to show $\frac{n}{2^n} \geq \frac{n+1}{2^{n+1}}$ $\forall n \in \mathbb{N}$.

Base Case: $\frac{1}{2} \geq \frac{2}{4}$; $\frac{1}{2} = \frac{2}{4}$, thus true.

Inductive Step: Assume $\frac{n}{2^n} \geq \frac{n+1}{2^{n+1}}$ $\forall n \in \mathbb{N}$. Need to show $\frac{n+1}{2^{n+1}} \geq \frac{n+2}{2^{n+2}}$

$$\frac{n+2}{2 \cdot 2 \cdot 2^n} < \frac{2n+2}{2 \cdot 2 \cdot 2^n} = \frac{n+1}{2 \cdot 2^n} = \frac{n+1}{2^{n+1}} = a_{n+1}.$$

Thus, $a_{n+2} \leq a_{n+1}$ for all n.

e) $\left\{\frac{n}{n+1}\right\} = \left(\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5} \ldots\right)$ is bounded, nondecreasing

$a_n \leq a_{n+1}$ for all n
$a_n = \frac{n}{n+1}$, $a_{n+1} = \frac{n+1}{n+2}$

Base Case: $\frac{1}{2} \leq \frac{2}{3}$ is true.
Inductive Step: Assume $a_n \leq a_{n+1}$. Need to show $a_{n+1} \leq a_{n+2}$
Proceed:
$$a_{n+1} = \frac{n+1}{n+2} = \frac{n}{n+2} + \frac{1}{n+2}$$

$(1-\frac{1}{2^n})a_n$

#3: If $a_1 = 1$ and $a_{n+1} = (1-2^{-n})a_n$, prove $\{a_n\}$ converges.

$(1, \frac{1}{2}, \frac{1}{2}(\frac{3}{4}), \frac{3}{8}(\frac{7}{8}), \frac{21}{64}(\frac{15}{16}), \ldots)$

The sequence is bounded by 0 and 1 (at least).

It also appears to be nonincreasing:

Claim: $a_n \geq a_{n+1}$ for all n (nonincreasing)

$a_n \geq (1-\frac{1}{2^n})a_n = a_{n+1}$

Base Case: $a_1 = 1 \geq (1-\frac{1}{2})1 = \frac{1}{2}$, thus true.

Inductive Step: Assume $a_n \geq a_{n+1}$ for all n by Inductive hypothesis.

Need to show $a_{n+1} \geq a_{n+2}$ for all n

Proceed:

$a_{n+2} = (1-\frac{1}{2^{n+1}})a_{n+1} = a_{n+1} - (a_{n+1})\frac{1}{2^{n+1}} < a_{n+1}$ which is true because $1/2^k$ is always a positive number for any $k \in \mathbb{N}$.

$\frac{1}{2^k} < 1$ for any $k \in \mathbb{N}$.

$2^k > 1$ is true for all k.

So, $(1-\frac{1}{2^k}) > 0$. Likewise, $1 > (1-\frac{1}{2^k})$.

This holds true when multiplying by numbers less than or equal to 1. Since $\{a_n\}$ is defined recursively by $a_{n+1} = (1-\frac{1}{2^n})a_n$ for all n, where $a_1 = 1$. Thus, all a_n will be between 0 and 1.

Thus, $\{a_n\}$ is both bounded and monotone. By MCT this implies convergence. So, $\{a_n\}$ converges. ∎

#s: 4.9

Homework 4, cont.

#4: Let $\{d_n\}$ be a sequence of 0's and 1's and define a seq. of numbers $\{a_n\}$ by:

$$a_n = d_1 2^{-1} + d_2 2^{-2} + d_3 2^{-3} + \dots d_n 2^{-n}.$$

Prove that this sequence converges to a number between 0 and 1.

For $\{d_{n_0}\}$ being a seq. of all 0s,
$a_{n_0} = 0$ for all n

For $\{d_{n_1}\}$ being a seq. of all 1s,
$a_{n_1} = 2^{-1} + 2^{-2} + \dots + 2^{-n}$.

So, $a_{n_0} \le a_n \le \sum_{k=1}^{n} 2^{-k} = a_{n_1}$

Then, $\sum_{k=1}^{n} 0 \cdot 2^{-k} \le \sum_{k=1}^{n} \begin{cases} 0 & \text{if } d_k = 0 \\ 2^{k} & \text{if } d_k = 1 \end{cases} \le \sum_{k=1}^{n} 2^{-k}$

So, $0 \le \sum_{k=1}^{n} d_k 2^{-k} \le \sum_{k=1}^{n} 2^{-k} < \sum_{k=1}^{\infty} 2^{-k} = \frac{a}{1-r} = \frac{1/2}{1-1/2} = 1$

Thus, $\{a_n\}$ is bounded.

Claim: $\{a_n\}$ is nondecreasing.

So, $a_{n+1} \ge a_n$ for all n,

$a_{n+1} = a_n + d_{n+1} 2^{-(n+1)} \ge a_n + 1 \cdot 2^{-(n+1)} > a_n$; $2^{-(n+1)} > 0$ "is positive"

Therefore, $\{a_n\}$ is bounded and monotone. So, by MCT, $\{a_n\}$ is convergent.

By Squeeze theorem, since $\{d_0\} \le \{a_n\} \le \{d_1\}$

where $d_0 \to 0$, $d_1 \to 1$

So, $0 \le \lim \{a_n\} \le 1$. ∎

2.4 #'s: 9

Homework 4, cont.

9: Prove $\lim \frac{2^n}{n} = \infty$.

Work: $\frac{2^n}{n}$; $\lim\left(\frac{n}{2^n}\right) \to 0$, So $\lim\left(\frac{2^n}{n}\right) \to \infty$ by 2.4.7 (a).

Proof:
Claim $2^n > n$ for all n.

Base Case: $2^1 > 1$; $2 > 1$ is true
Inductive Step: Assume $2^n > n$ for all n. Need to show $2^{n+1} > n+1$.
Proceed by induction.

$2^{n+1} = 2 \cdot 2^n > 2 \cdot n \geq n+1$.

Claim $2n \geq n+1$ for all n.
Base Case: $2(1) \geq (1+1)$; $2 = 2$ is true
Inductive Step: Assume $2n \geq n+1$ for all n. Need to show $2(n+1) \geq (n+1)+1$ for all n.
$2(n+1) = 2n+2 = 2n+1+1 > n+1+1 = (n+1)+1$.
So, $2n \geq n+1$ for all n

And thus $2^n > n$ for all n. □

Thus, $\frac{n}{2^n} < \frac{n}{n} = 1$. $\frac{0}{2^n} < \frac{n}{2^n}$, so $0 < \frac{n}{2^n}$.

Therefore, $\frac{n}{2^n}$ is bounded.

Claim $\frac{n}{2^n}$ is monotone nonincreasing: $a_{n+1} \leq a_n$

$a_{n+1} = \frac{n+1}{2 \cdot 2^n}$, $a_n = \frac{n}{2^n}$

Base Case: $P(1)$: $\frac{1}{2} \leq \frac{1}{2}$ is true
Inductive Step: Assume $a_{n+1} \leq a_n$, need to show $a_{n+2} \leq a_{n+1}$.
Proceed: $a_{n+2} = \frac{n+2}{2 \cdot 2 \cdot 2^n} < \frac{2n+2}{2 \cdot 2 \cdot 2^n} = \frac{n+1}{2 \cdot 2^n} \leq a_{n+1}$. $\{a_n\}$ is monotone.

Therefore, $\{a_n\}$ is monotone & bounded. It is therefore convergent. MCT.

Claim $a_n \to 0$; $\frac{n}{2^n} < \frac{n}{1} < \varepsilon$; So, for $n > N = \varepsilon$
$\frac{n}{2^n} \to 0$.

Scratch

$\frac{2^n}{n} > n$ for all n

$\frac{2^{n+1}}{n+1} > n+1$?, $2 \cdot 2^n > n^2+2n+1$

$2^n+2^n > n^2+2n+1$ show $2^n \geq n^2+1$

$$\frac{2 \cdot 2^n}{n+1} = \frac{2^n}{n+1} + \frac{2^n}{n+1} \geq \frac{n}{n+1}$$

$$2^{n+1} \geq (n+1)^2+1 = n^2+2n+1$$

True by inductive Hypothesis

$2^n > n^2$ for $n \geq 4$

$2 \cdot 2^n >$

$n^2 \geq 2n+1$ for $n \geq 4$

$(n+1)^2 \geq 2(n+1)+1$

$n^2+2n+1 \geq 2n+1+2n+1 = 2n+n+n+2 > 2n+1+2$ true

By Binomial Theorem

$\lim \frac{2^n}{n} = \infty$. Proof: Use Bin Thm on 2^n.

$$\frac{1}{n}\left[2^n = (1+1)^n \underset{b=1}{} \Rightarrow 1+n+\frac{n(n-1)}{2}+\dots+1^n\right] = \frac{1}{n}+1+\frac{n-1}{2}+\dots+\frac{1}{n}^n$$

$\geq n-1 > n$ which is unbounded by Arch Prop

So, by (d) of Thm 2.4.7

$$\lim (n) < \lim \left(\frac{2^n}{n}\right)$$

So, $\lim \frac{2^n}{n} \to \infty$. ∎